SOLAR ENERGY PROPOSAL · ข้อเสนอพลังงานแสงอาทิตย์

# Solar Power for Kaniel Villas

Independent 27.8 kWp solar system for each of 3 luxury villas — Koh Phangan, Surat Thani

ระบบโซลาร์เซลล์อิสระ 27.8 kWp สำหรับวิลล่า 3 หลัง — เกาะพะงัน สุราษฎร์ธานี

**฿629,938**
Annual Savings / ประหยัดต่อปี
3 villas × ฿209,979/yr at ฿6/kWh PEA rate

**3.8 yrs**
Payback Period / คืนทุน

**104,990**
kWh/year Production

**25 yrs**
Panel Warranty

| Prepared for | Proposal Date | Reference | Valid Until |
|---|---|---|---|
| **Kaniel Neighborhood Villas** | **14 March 2026** | **TM-2026-003** | **13 April 2026 (30 days)** |

YOUR PROJECT · โปรเจกต์ของคุณ

# Three Villas, Three Independent Systems

Each villa receives its own complete solar system — independent inverter, independent metering, with optional battery backup.

วิลล่าแต่ละหลังได้รับระบบโซลาร์ครบชุด — อินเวอร์เตอร์แยก มิเตอร์แยก พร้อมแบตเตอรี่สำรอง

Aerial Rendering — 144 Panels · 83.5 kWp across 3 Villas

Huawei SUN2000-25KTL + LUNA2000-10kWh Battery

### 🏡 PROPERTY DETAILS / รายละเอียดโครงการ

| | |
|---|---|
| Property | **Kaniel Neighborhood — 3 Luxury Villas** |
| Location | **Koh Phangan, Surat Thani** |
| Villas | **3 identical villas with pools** |
| Consumption | **~2,000 kWh/month per villa** |

### ☀️ ROOF ASSESSMENT / ประเมินหลังคา

| | |
|---|---|
| Roof Type | **Gable — Metal Sheet** |
| Roof Size | **~18.3m × 13.2m per villa** |
| Usable Slope | **~140 m² (south-facing)** |
| Roof Pitch | **~13° (optimal for 10°N latitude)** |

☀️

**146%**

Solar covers more than your consumption

⚡

**3 × Independent**

Each villa runs its own system

🔋

**8h Backup**

Emergency power with battery option

PER-VILLA SYSTEM · ระบบแต่ละวิลล่า

# Each Villa Gets a Complete System

Fully independent — if one system needs maintenance, the other two keep running.

เป็นอิสระจากกัน — หากระบบหนึ่งต้องซ่อมบำรุง อีกสองระบบยังทำงานต่อ

**48**

**LONGi 580W Panels**

3 rows × 16 columns · portrait

27.8 kWp per villa

**1**

**Huawei Inverter**

SUN2000-25KTL-M5

25 kW · 4 MPPT · 98.6% eff.

**10** kWh

**Battery (Option C)**

LUNA2000 · 5kW output

~8 hours emergency backup

### Monthly Production vs Consumption

| | |
|---|---|
| Solar Production | 2,916 kWh/mo |
| Villa Consumption | ~2,000 kWh/mo |
| Coverage | 146% ✓ |
| Monthly Surplus | ~916 kWh |

### Emergency Backup Loads (Option C)

| | |
|---|---|
| Air Conditioning (1 unit) | 1.20 kW |
| WiFi Router + Modem | 0.05 kW |
| LED Lighting (5 points) | 0.05 kW |
| Water Pump | 0.75 kW |
| **Total Peak Load** | **2.05 kW** |
| **Backup Duration** | **~8 hours** |

CHOOSE YOUR PLAN · เลือกแผนของคุณ

# Three Options, One Goal: Lower Bills

All prices shown per villa. All 3 villas use identical equipment.

ราคาแสดงต่อวิลล่า วิลล่าทั้ง 3 หลังใช้อุปกรณ์เหมือนกัน

## Option A: PPA

Zero upfront — we install, you save / ไม่ต้องจ่ายล่วงหน้า

**฿0 Upfront**

pay only ฿4.50/kWh (25% less than PEA)

**What is PPA?** We install solar on your roof at zero cost. You simply buy the electricity at ฿4.50/kWh — 25% cheaper than PEA's ฿6/kWh rate.

PPA คืออะไร? เราติดตั้งโซลาร์ฟรี คุณเพียงซื้อไฟฟ้าในราคา ฿4.50/kWh — ถูกกว่า PEA 25%

- ฿0 installation — TM Energy pays for everything
- 25% cheaper than PEA grid rate
- Free maintenance & monitoring for 20 years
- Free panel cleaning 2× per year
- System insured against natural disasters
- Start saving from Day 1 — zero risk

Per Villa Monthly Savings

★ RECOMMENDED / แนะนำ

## Option B: EPC

Own your system outright / เป็นเจ้าของระบบ

**฿807,360**

per villa / ต่อวิลล่า (฿29/Wp)

**3 Villas Total: ฿2,422,080**

30% deposit · 40% on delivery · 30% on commissioning

- 48 panels + inverter per villa — fully installed
- PEA license & grid connection included
- 25-year panel warranty
- 10-year inverter warranty
- 24/7 monitoring (1 year free)
- System insured against natural disasters
- You own the system — all savings are yours

Per Villa Annual Savings

**฿209,979**

## Option C: EPC + Battery

Solar + emergency backup / โซลาร์ + สำรองไฟฉุกเฉิน

**฿947,360**

per villa / ต่อวิลล่า (solar + 10 kWh battery)

**3 Villas Total: ฿2,842,080**

30% deposit · 40% on delivery · 30% on commissioning

- Everything in Option B, plus:
- Huawei LUNA2000-10kWh battery per villa
- ~8 hours emergency blackout protection
- Covers AC + WiFi + lights + water pump
- 10-year battery warranty
- Smart energy management via Huawei app

Per Villa Annual Savings

**฿209,979**

**~฿2,500/mo**

Savings start **immediately** / เริ่มประหยัดทันที

Payback: **3.8 years** / คืนทุน 3.8 ปี

Payback: **4.5 years** · + peace of mind

SIDE-BY-SIDE COMPARISON · เปรียบเทียบตัวเลือก

# Compare All Three Options

Prices shown per villa. All 3 villas are identical systems.

ราคาแสดงต่อวิลล่า ระบบเหมือนกันทั้ง 3 หลัง

| | ☀️ Option A: PPA | ⚡ Option B: EPC ★ | 🔋 Option C: EPC+Battery |
|---|---|---|---|
| Upfront Cost (per villa) | **฿0** | **฿807,360** | **฿947,360** |
| 3 Villas Total | **฿0** | **฿2,422,080** | **฿2,842,080** |
| Monthly Payment | ~฿13,125/villa<br>฿4.50 × kWh | ฿0 after purchase | ฿0 after purchase |
| Annual Savings (per villa) | **~฿30,000** | **฿209,979** | **฿209,979** |
| Payback Period | **Immediate** | **3.8 years** | **4.5 years** |
| Blackout Protection | ❌ Daytime only | ❌ Daytime only | ✅ **~8 hours** |
| Ownership | TM Energy<br>(buy after 20 yrs) | You own it ✓ | You own it ✓ |

SYSTEM DETAILS · รายละเอียดระบบ

# Premium Equipment, Built to Last

Each villa receives identical Tier 1 equipment — designed for Thailand's tropical climate.

วิลล่าแต่ละหลังใช้อุปกรณ์ Tier 1 เหมือนกัน ออกแบบสำหรับเขตร้อน

### ☀️ Solar Panels — LONGi Hi-MO X6

| | |
|---|---|
| Model | LR5-72HGD 580W |
| Per Villa | 48 panels · 27.8 kWp |
| Total (3 villas) | 144 panels · 83.5 kWp |
| Layout | 3 rows × 16 cols (portrait) |
| Cell Technology | N-type TOPCon · 22.3% eff. |
| Warranty | 25-year product · 30-year perf. |

### ⚡ Inverter — Huawei SUN2000-25KTL-M5

| | |
|---|---|
| Per Villa | 1 unit · 25 kW |
| Efficiency | 98.6% · IP66 outdoor rated |
| MPPT | 4 trackers · AI-powered |
| Monitoring | Huawei FusionSolar App |
| Warranty | 10 years |

### 🔋 Battery — Huawei LUNA2000-10kWh (Option C)

| | |
|---|---|
| Per Villa | 1 unit · 10 kWh |
| Output Power | 5 kW (continuous) |
| Backup Duration | ~8 hours (AC+WiFi+lights+pump) |
| Depth of Discharge | 100% DoD |
| Cycle Life | 6,000+ cycles |
| Warranty | 10 years |
| Expandable | Up to 30 kWh (add modules later) |

### 🔧 Installation Includes

| | |
|---|---|
| Mounting | No-drill roof clamps |
| Wind Rating | Up to 60 m/s (typhoon rated) |
| Protection | Surge + circuit breakers |
| PEA License | Application handled by us |

YOUR SAVINGS · เงินที่คุณประหยัดได้

# See How Much You'll Save

| ฿629,938 | ฿6.3M | 3.8 | 52.5t |
|---|---|---|---|
| Annual Savings (3 villas)<br>ประหยัดต่อปี | 10-Year Savings<br>ประหยัด 10 ปี | Years to Payback<br>ปีที่คืนทุน | $CO_2$ Saved/Year<br>ลด $CO_2$ ต่อปี |

HOW IT WORKS · ขั้นตอนการดำเนินงาน

# From Approval to Power in 6–8 Weeks

**WEEK 1**

**Survey & Design**

Site measurement, system design for each villa

**WEEK 2–3**

**Permits & Procurement**

3× PEA licenses, equipment order

**WEEK 4–6**

**Installation**

Mount panels, install inverters + batteries on all 3 villas

**WEEK 7–8**

**Go Live ✓**

Testing, PEA inspection, grid connection, monitoring setup

WARRANTIES & SUPPORT · การรับประกัน

# Protected for Decades

**25** years

**Solar Panel Warranty**

LONGi product warranty
30-year performance guarantee

**10** years

**Inverter Warranty**

Huawei manufacturer warranty

**10** years

**Battery Warranty**

LUNA2000 · 6,000+ cycles

**24** months

**Workmanship**

TM Energy installation warranty

**24/7**

**Remote Monitoring**

FusionSolar App

**Local**

**On-Island Team**

Same-day response

# Ready to Start Saving?

## พร้อมเริ่มประหยัดแล้วหรือยัง?

This proposal is valid for 30 days. Let's discuss your preferred option.

💬 **WhatsApp Us Now** | 📞 **062-348-0119**

**1.** Choose your option **2.** Sign agreement **3.** We start within 1 week

ABOUT US · เกี่ยวกับเรา

# About TM Energy

TM Energy delivers professional solar energy solutions across Koh Phangan and the Gulf of Thailand. We design, install, and maintain high-performance solar systems — with a local team that provides same-day service.

**Powered by TM Developers** — Koh Phangan's trusted construction company.

# Service Agreement

## SELECTED OPTION

- ○ Option A: PPA — ฿0 Upfront
- ○ Option B: EPC — ฿807,360/villa
- ○ Option C: EPC+Battery — ฿947,360/villa

## KEY TERMS

- System: 3× independent 27.8 kWp solar installations (48× LONGi Hi-MO X6 580W panels each)
- Inverters: 3× Huawei SUN2000-25KTL-M5 (25 kW each)
- If Option C: + 3× Huawei LUNA2000-10kWh battery (10 kWh each)
- Installation timeline: 6–8 weeks from signing
- Warranty: 25-year panels, 10-year inverters, 10-year battery, 24-month workmanship

▶ **Terms & Conditions**

| CLIENT | TM ENERGY |
|---|---|
| _____ | _____ |
| FULL NAME | FULL NAME |
| DATE | DATE |

TM ENERGY

Powered by TM Developers — Koh Phangan

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